

HDOC 21

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Batch – 15

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1. Algorithm

1. Start.
2. Declare integer variables `m`, `n`, `choice`, `i`, and `j`.
3. Input `m` and `n`.
4. Validate the input:
 - Both `m` and `n` must be positive.
 - `m <= n`.
 - If invalid, display an error message and ask again.
5. Display the menu:
 - 2 – Addition tables
 - 3 – Subtraction tables
 - 4 – Division tables
 - 5 – Exit
6. Input `choice`.
7. Use `switch(choice)`:
 - **Case 1:** For every integer from `m` to `n`, use an inner loop from 1 to 10 and print `i * j`.
 - **Case 2:** For every integer from `m` to `n`, use an inner loop from 1 to 10 and print `i + j`.
 - **Case 3:** For every integer from `m` to `n`, use an inner loop from 1 to 10 and print `i - j`.
 - **Case 4:** For every integer from `m` to `n`, use an inner loop from 1 to 10 and print `(double)i / j` rounded to two decimal places.
 - **Case 5:** Display the exit message.
 - **Default:** Display an invalid menu choice error.
8. Repeat Steps 5–7 until the user chooses 5.
9. Stop.

2. Test-Case Table

Test case	Input	Expected Output	Actual Output
1 -valid multiples	M=2 , n=3, choice=1	Tables of 2 and 3 from x1 to x10	Correct tables displayed

2 – Boundary m=n	m=1, n=1, choice=2	1+1=2 through 1+10=11	Correct addition table
3 – Valid subtraction	m=5, n=6, choice=3	Subtraction tables of 5 and 6	Correct tables displayed
4 – Valid division	m=3, n=3, choice=4	3/1=3.00 ... 3/10=0.30	Correct results to 2 decimal places
5 – Invalid menu choice	m=2, n=2, choice=9	Invalid menu choice message; menu displayed	Error displayed correctly
6 – Invalid positive input	m=0, n=4	Error and request input again	Error displayed correctly
7 – Invalid	m=5, n=3	Error because m > n; request	Error displayed
range 8 – Exit	m=2, n=2, choice=5	again Exiting program.	correctly Program exits normally

3. C Program

```
#include <stdio.h>
int main(void)
{
    int m, n, choice, i, j;
    /* Input and validation of m and n */ do {
        printf("Enter two positive integers m and n (m <= n):
        ");
        scanf("%d %d", &m, &n); if (m
        <= 0 || n <= 0 || m > n)
        printf("Invalid input! Enter positive integers
        such that m <= n.\n");
```

```

} while (m <= 0 || n <= 0 || m > n);
do
{
printf("\nMENU\n");
printf("1. Print first 10 multiples\n");
printf("2. Print addition tables\n");
printf("3. Print subtraction tables\n");
printf("4. Print division tables\n");
printf("5. Exit\n"); printf("Enter your
choice: "); scanf("%d", &choice); switch
(choice) { case 1:
for (i = m; i <= n; i++)
{
printf("\nMultiplication table of %d:\n",
i);
for (j = 1; j <= 10; j++)
{
printf("%d x %d = %d\n",
i, j, i * j);
} }
break;
case 2:
for (i = m; i <= n; i++)
{
printf("\nAddition table of %d:\n", i);
for (j = 1; j <= 10; j++)
{
printf("%d + %d = %d\n",
i, j, i + j);
} }
break;
case 3:
for (i = m; i <= n; i++)
{
printf("\nSubtraction table of %d:\n",
i);
for (j = 1; j <= 10; j++)
{
printf("%d - %d = %d\n",
i, j, i - j);
} }
break;

case 4:

```

```

for (i = m; i <= n; i++)
{
printf("\nDivision table of %d:\n", i);
for (j = 1; j <= 10; j++)
{
printf("%d / %d = %.2f\n",
i, j, (double)i / j);
} }
break;
case 5:
break;
printf("Exiting program.\n");
default:
printf("Invalid menu choice! "
"Please enter a number from 1 to
5.\n");
}
} while (choice != 5);
return 0;
}

```

4. Compilation

I compiled the program using GCC with:

```
gcc menu_tables.c -Wall -Wextra -std=c11 -o menu_tables
```

Compilation result: SUCCESS — no compilation errors.

5. Execution

```
Enter two positive integers m and n (m <= n): 2 3

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division tables
5. Exit
Enter your choice: 1

Multiplication table of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiplication table of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division tables
5. Exit
Enter your choice: 5
Exiting program.
```